DRUG NAME: Atenolol
Class: Beta-blocker

Indications:

- Hypertension
- Angina pectoris

Contraindications:

- Severe bradycardia
- Bronchial asthma

Drug Interactions:

- Verapamil → Increased risk of severe bradycardia and heart block
- NSAIDs → Reduced antihypertensive effect

Common Side Effects:

- Fatigue
- Dizziness

DRUG NAME: Amlodipine
Class: Calcium channel blocker

Indications:

- Hypertension
- Chronic stable angina

Contraindications:

- Severe hypotension
- Cardiogenic shock

Drug Interactions:

- Simvastatin → Increased risk of myopathy
- Rifampicin → Reduced antihypertensive effect

Common Side Effects:

- Peripheral edema
- Headache

DRUG NAME: Metformin
Class: Biguanide antidiabetic

Indications:

- Type 2 diabetes mellitus
- Prediabetes

Contraindications:

- Severe renal impairment
- Metabolic acidosis

Drug Interactions:

- Cimetidine → Increased risk of lactic acidosis
- Iodinated contrast media → Increased risk of renal impairment and lactic acidosis

Common Side Effects:

- Diarrhea
- Nausea

DRUG NAME: Artemether-Lumefantrine
Class: Artemisinin-based combination antimalarial

Indications:

- Uncomplicated Plasmodium falciparum malaria
- Mixed malaria infections

Contraindications:

- Hypersensitivity to artemisinin derivatives
- Severe hepatic impairment

Drug Interactions:

- Rifampicin → Reduced antimalarial efficacy
- Efavirenz → Reduced lumefantrine levels

Common Side Effects:

- Headache
- Dizziness

DRUG NAME: Paracetamol
Class: Analgesic and antipyretic

Indications:

- Mild to moderate pain
- Fever

Contraindications:

- Severe hepatic impairment
- Hypersensitivity to paracetamol

Drug Interactions:

- Warfarin → Increased risk of bleeding with prolonged use
- Isoniazid → Increased risk of hepatotoxicity

Common Side Effects:

- Nausea
- Rash

DRUG NAME: Ciprofloxacin
Class: Fluoroquinolone antibiotic

Indications:

- Urinary tract infections
- Gastrointestinal infections

Contraindications:

- Hypersensitivity to fluoroquinolones
- History of tendon disorders related to quinolones

Drug Interactions:

- Theophylline → Increased risk of toxicity
- Antacids → Reduced ciprofloxacin absorption

Common Side Effects:

- Nausea
- Diarrhea

DRUG NAME: Omeprazole
Class: Proton pump inhibitor

Indications:

- Peptic ulcer disease
- Gastroesophageal reflux disease

Contraindications:

- Hypersensitivity to proton pump inhibitors
- Concomitant use with nelfinavir

Drug Interactions:

- Clopidogrel → Reduced antiplatelet effect
- Warfarin → Increased risk of bleeding

Common Side Effects:

- Headache
- Abdominal pain

DRUG NAME: Aspirin
Class: Antiplatelet and NSAID

Indications:

- Secondary prevention of myocardial infarction
- Mild pain and fever

Contraindications:

- Active peptic ulcer disease
- Bleeding disorders

Drug Interactions:

- Warfarin → Increased bleeding risk
- Ibuprofen → Reduced antiplatelet effect
- Gastric irritation
- Bleeding

DRUG NAME: Lisinopril
Class: ACE inhibitor

Indications:

- Hypertension
- Heart failure

Contraindications:

- History of angioedema
- Pregnancy

Drug Interactions:

- Spironolactone → Increased risk of hyperkalemia
- NSAIDs → Reduced antihypertensive effect and renal impairment

Common Side Effects:

- Dry cough
- Hyperkalemia

DRUG NAME: Verapamil
Class: Calcium channel blocker

Indications:

- Hypertension
- Supraventricular tachycardia

Contraindications:

- Severe left ventricular dysfunction
- Second or third-degree AV block

Drug Interactions:

- Beta-blockers → Increased risk of bradycardia and heart block
- Digoxin → Increased digoxin levels

Common Side Effects:

- Constipation
- Bradycardia

DRUG NAME: Salbutamol
Class: Short-acting beta2-agonist

Indications:

- Acute asthma
- Bronchospasm

Contraindications:

- Hypersensitivity to salbutamol
- Severe tachyarrhythmias

Drug Interactions:

- Beta-blockers → Reduced bronchodilatory effect
- Diuretics → Increased risk of hypokalemia

Common Side Effects:

- Tremor
- Palpitations

DRUG NAME: Losartan
Class: Angiotensin II receptor blocker

Indications:

- Hypertension
- Diabetic nephropathy

Contraindications:

- Pregnancy
- Bilateral renal artery stenosis

Drug Interactions:

- Potassium supplements → Increased risk of hyperkalemia
- NSAIDs → Reduced antihypertensive effect

Common Side Effects:

- Dizziness
- Hyperkalemia

DRUG NAME: Hydrochlorothiazide
Class: Thiazide diuretic

Indications:

- Hypertension
- Edema

Contraindications:

- Anuria
- Severe electrolyte imbalance

Drug Interactions:

- Lithium → Increased lithium toxicity
- Digoxin → Increased risk of arrhythmias due to hypokalemia

Common Side Effects:

- Hypokalemia
- Hyperuricemia

DRUG NAME: Insulin (Regular)
Class: Short-acting insulin

Indications:

- Type 1 diabetes mellitus
- Type 2 diabetes mellitus

Contraindications:

- Hypoglycemia
- Hypersensitivity to insulin

Drug Interactions:

- Beta-blockers → Masked hypoglycemia symptoms
- Corticosteroids → Increased blood glucose levels

Common Side Effects:

- Hypoglycemia
- Weight gain

DRUG NAME: Glibenclamide
Class: Sulfonylurea antidiabetic

Indications:

- Type 2 diabetes mellitus
- Adjunct to diet and exercise

Contraindications:

- Type 1 diabetes mellitus
- Severe hepatic impairment

Drug Interactions:

- Fluconazole → Increased risk of hypoglycemia
- Beta-blockers → Masked hypoglycemia symptoms

Common Side Effects:

- Hypoglycemia
- Weight gain

DRUG NAME: Azithromycin
Class: Macrolide antibiotic

Indications:

- Respiratory tract infections
- Sexually transmitted infections

Contraindications:

- Hypersensitivity to macrolides
- History of cholestatic jaundice due to azithromycin

Drug Interactions:

- Warfarin → Increased bleeding risk
- Antacids → Reduced azithromycin absorption

Common Side Effects:

- Diarrhea
- Nausea

DRUG NAME: Ceftriaxone
Class: Third-generation cephalosporin

Indications:

- Severe bacterial infections
- Meningitis

Contraindications:

- Hypersensitivity to cephalosporins
- Neonates with hyperbilirubinemia

Drug Interactions:

- Calcium-containing solutions → Risk of precipitation
- Warfarin → Increased bleeding risk

Common Side Effects:

- Diarrhea
- Injection site pain

DRUG NAME: Amoxicillin-Clavulanate
Class: Beta-lactam antibiotic combination

Indications:

- Respiratory tract infections
- Skin and soft tissue infections

Contraindications:

- History of penicillin allergy
- History of cholestatic jaundice due to amoxicillin-clavulanate

Drug Interactions:

- Warfarin → Increased bleeding risk
- Methotrexate → Increased methotrexate toxicity

Common Side Effects:

- Diarrhea
- Rash

DRUG NAME: Diclofenac
Class: Nonsteroidal anti-inflammatory drug

Indications:

- Musculoskeletal pain
- Inflammatory conditions

Contraindications:

- Active peptic ulcer disease
- Severe heart failure

Drug Interactions:

- ACE inhibitors → Reduced antihypertensive effect
- Warfarin → Increased bleeding risk

Common Side Effects:

- Gastric irritation
- Elevated liver enzymes

DRUG NAME: Ibuprofen
Class: Nonsteroidal anti-inflammatory drug

Indications:

- Mild to moderate pain
- Fever

Contraindications:

- Active gastrointestinal bleeding
- Severe renal impairment

Drug Interactions:

- Aspirin → Reduced antiplatelet effect
- Antihypertensives → Reduced antihypertensive effect

Common Side Effects:

- Dyspepsia
- Nausea

DRUG NAME: Nitroglycerin
Class: Nitrate vasodilator

Indications:

- Acute angina
- Heart failure

Contraindications:

- Hypotension
- Concomitant use with phosphodiesterase-5 inhibitors

Drug Interactions:

- Sildenafil → Severe hypotension
- Alcohol → Enhanced hypotensive effect

Common Side Effects:

- Headache
- Flushing

DRUG NAME: Spironolactone
Class: Potassium-sparing diuretic

Indications:

- Heart failure
- Resistant hypertension

Contraindications:

- Hyperkalemia
- Severe renal impairment

Drug Interactions:

- ACE inhibitors → Increased risk of hyperkalemia
- Potassium supplements → Increased risk of hyperkalemia

Common Side Effects:

- Hyperkalemia
- Gynecomastia

DRUG NAME: Furosemide
Class: Loop diuretic

Indications:

- Edema
- Hypertension

Contraindications:

- Anuria
- Severe electrolyte depletion

Drug Interactions:

- Aminoglycosides → Increased ototoxicity
- Digoxin → Increased risk of arrhythmias due to hypokalemia

Common Side Effects:

- Hypokalemia
- Dehydration

DRUG NAME: Clopidogrel
Class: Antiplatelet agent

Indications:

- Acute coronary syndrome
- Secondary prevention of stroke

Contraindications:

- Active bleeding
- Severe hepatic impairment

Drug Interactions:

- Omeprazole → Reduced antiplatelet effect
- Warfarin → Increased bleeding risk

Common Side Effects:

- Bleeding
- Bruising

DRUG NAME: Warfarin
Class: Vitamin K antagonist anticoagulant

Indications:

- Atrial fibrillation
- Deep vein thrombosis

Contraindications:

- Active bleeding
- Pregnancy

Drug Interactions:

- Metronidazole → Increased INR and bleeding risk
- Rifampicin → Reduced anticoagulant effect

Common Side Effects:

- Bleeding
- Skin necrosis

DRUG NAME: Atorvastatin
Class: HMG-CoA reductase inhibitor

Indications:

- Hyperlipidemia
- Prevention of cardiovascular events

Contraindications:

- Active liver disease
- Pregnancy

Drug Interactions:

- Clarithromycin → Increased risk of myopathy
- Cyclosporine → Increased statin levels

Common Side Effects:

- Myalgia
- Elevated liver enzymes

DRUG NAME: Metronidazole
Class: Nitroimidazole antibiotic

Indications:

- Anaerobic infections
- Amoebiasis

Contraindications:

- First trimester of pregnancy
- Hypersensitivity to nitroimidazoles

Drug Interactions:

- Warfarin → Increased INR and bleeding risk
- Alcohol → Disulfiram-like reaction

Common Side Effects:

- Metallic taste
- Nausea

DRUG NAME: Ranitidine
Class: H2 receptor antagonist

Indications:

- Peptic ulcer disease
- Gastroesophageal reflux disease

Contraindications:

- Hypersensitivity to ranitidine
- Acute porphyria

Drug Interactions:

- Warfarin → Altered anticoagulant effect
- Ketoconazole → Reduced antifungal absorption

Common Side Effects:

- Headache
- Constipation

DRUG NAME: Prednisolone
Class: Corticosteroid

Indications:

- Inflammatory conditions
- Severe asthma

Contraindications:

- Systemic fungal infections
- Live vaccines

Drug Interactions:

- NSAIDs → Increased risk of gastrointestinal bleeding
- Insulin → Increased blood glucose levels

Common Side Effects:

- Hyperglycemia
- Weight gain

DRUG NAME: Artesunate
Class: Artemisinin derivative antimalarial

Indications:

- Severe malaria
- Complicated Plasmodium falciparum malaria

Contraindications:

- Hypersensitivity to artemisinin derivatives
- First trimester of pregnancy unless benefits outweigh risks

Drug Interactions:

- Rifampicin → Reduced antimalarial efficacy
- Efavirenz → Reduced artesunate levels

Common Side Effects:

- Dizziness
- Transient neutropenia